

DECT NR+:Unveiling the Essentials of a new non-cellular 5G Standard for Verticals

1st M. Dolores Pérez-Guirao
Communication Technology
Ostfalia HAW
Wolfenbüttel, Germany
m.perez@ostfalia.de

2nd Thomas Weisshaupt
Head of D-A-CH Region
Wirepas Oy
City, Finland
thomas.weisshaupt@wirepas.com

3rd Andreas Wilzeck
Spectrum Policy and Standards
Sennheiser electronic GmbH & Co. KG
Wedermark, Germany
andreas.wilzeck@sennheiser.com

Abstract—DECT-2020 NR is a young radio standard developed by ETSI, which aims to provide a lean but powerful technical foundation for industrial IoT applications (IIoT) such as e.g. Industry 4.0, media content production, mission-critical or building automation and monitoring. The DECT Forum markets DECT-2020 NR as DECT NR+. In the fall of 2021, ITU-R adopted DECT-2020 NR as part of the IMT-2020 family of standards, also known as the 5th generation (5G) of wireless access technology. Driving underlying design objective of DECT-2020 NR has been to create a wireless communication technology, which can be deployed by anyone, be operated by anybody and used anywhere opening in that sense extensive opportunities for sustainable, independent or cooperative wireless connectivity solutions, especially for medium-sized industries. This paper provides an overview of the essentials of the state of the art in DECT NR+ represented by the ETSI TC DECT-2020 NR technical specifications. We describe the fundamental concepts of DECT-2020 NR, highlighting how application and solution development is facilitated, and propose various possible innovation paths influencing future standardization and spectrum policy concepts for local area private industrial networks. In addition, we point out options for governance models that can lead to democratisation of 5G and IIoT.

Index Terms—DECT-2020 NR, DECT NR+, IMT-2020, 5G, private networks, spectrum policy, governance

I. INTRODUCTION

DECT is a long-standing and successful wireless technology, which has its most recognized and widespread application area in the cordless telephony. However, many other diverse applications ranging from audio microphones and headsets to smart home and smart building apply DECT connectivity solutions.

ETSI released the first DECT Standard in 1992. Since then, the standardization process has never stop re-shaping itself, adding innovative features and targeting new markets. For instance, in 2013 the DECT Ultra Low Energy (ULE) standard was added to the DECT family to target IoT centric markets like smart home and smart building. In fall 2021, the latest DECT specification, known as DECT-2020 NR, was recognized by ITU as a 5G (i.e. IMT-2020 conform) technology, with focus on ultra-reliable low latency and massive machine type communications, i.e. URLLC and mMTC, respectively. Since then, DECT-2020 NR has enriched the IMT-2020 standards landscape as the only non-cellular 5G standard.

The rest of the paper is organized as follows. The relationship between DECT-2020 NR and IMT-2020, i.e. the fifth-generation of mobile communication standards, is explained in Section II. Section III outlines the fundamental concepts, technical parameters and network architecture of DECT-2020 NR, with particular focus on its capacity to deploy scalable, self-organizing networks. Section IV exposes some potential vertical markets, and their requirements, that can stand to benefit from DECT-2020 NR. Future work directions in spectrum policy and governance models are derived in Section V for the highlighted vertical markets. Section V concludes the paper.

II. DECT-2020 NR IS A NON-CELLULAR 5G STANDARD

DECT-2020 NR technology underwent the international mobile telecommunication (IMT) evaluation process as a set of radio interface technologies (SRIT) known as DECT 5G-SRIT. Fig.1 illustrates DECT 5G-SRIT consisting of the two radio interface technologies (RIT) DECT-2020 NR and 3GPP NR. DECT-2020 NR RIT component comprises Releases 1 and beyond, while 3GPP NR RIT component comprises Releases 15 and beyond. In order to include DECT 5G-SRIT as an IMT-2020 technology, ITU-R appointed various independent assessment groups to evaluate the performance of the RIT components, specially DECT-2020 NR, against the IMT-2020 requirements [1]–[3].

The IMT evaluation process concluded in fall 2021 and Recommendation ITU-R M.2150-1 [4] confirmed that the DECT-5G-SRIT fulfils all technical performance requirements in all five selected test environments and that the component RIT DECT-2020 NR fulfils the technical performance requirements in two selected test environments: Urban Macro – Ultra Reliable Low Latency Communication (URLLC) and Urban Macro massive Machine Type Communication (mMTC). It is worth noting that the DECT-2020 NR RIT can also handle reliable, broadband communications at high data rates (several Gbit/s), provided sufficient radio spectrum is available.

Certainly, a relevant feature of the legacy DECT technology is its license-exempted operation in radio bands around 1,9 GHz. For the European Union the decision ERC/DEC/(94)03 [5] dedicates 1880-1900 MHz specifically to DECT, so that DECT operation shall have priority over other radio systems and be protected from harmful interference in the designated

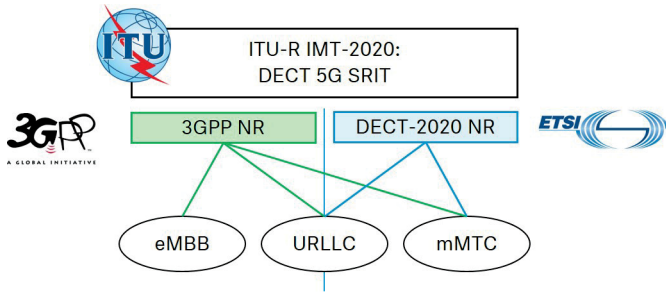


Fig. 1. DECT-2020 has been recognized by ITU-R (WP5D) as IMT-2020 technology.

band. In addition to the legacy DECT core band [1880 - 1900/1920] MHz, the DECT-2020 NR standard is designed for all IMT TDD frequencies below 6 GHz, thus expanding far beyond the legacy core band.

III. FUNDAMENTALS OF DECT NR+

DECT NR+ is a local, self-organizing radio network technology that supports numerous network topologies and features automatic interference management that eliminates the need for manual network or frequency planning. In this section, we present the fundamental concepts of DECT NR+ design and basic terminologies, with a comparison of the principal differences between cellular 3GPP NR and DECT NR+ given in Table I.

A. Design Objectives: Anywhere, Anytime, Anyone

DECT-2020 NR was deliberately kept agnostic to use cases and business models of manufacturers, users or operators but taking into account that the possibility to build and operate own, local networks is indispensable for many vertical industries (e.g. Industry 4.0, content production, utility and smart cities).

The ultimate goal of ETSI by the design of the DECT-2020 NR technical specifications was to offer a local but scalable, self-contained wireless connectivity solution that could be self-deployed and self-operated by industrial users (anyone) anywhere and anytime. Concretely, this means that DECT-2020 NR can be used without separate infrastructure or in conjunction with any IP-enabled Wide Area Network (WAN) infrastructure (e.g., fiber, satellite, or public cellular network); it also allows the use of proprietary (i.e. non-IP) network or application protocols without compromising security.

This unique set of circumstances offers significant benefits for industries that require reliable service for production-relevant and mission critical processes but not necessarily the custody or intermediation of a commercial network operator. Industrial and process automation, media content production, smart building, and PPDR are examples of industries that attach top priority to the possibility to set up and operate their own mission critical private local networks and can stand to benefit from DECT NR+ unique connectivity proposition.

TABLE I
COMPARISON BETWEEN 3GPP NR AND DECT NR+

	3GPP NR	DECT NR+
Deployment	Public Network (PLMN) Non-Public Network (NPN)	Anywhere, Anytime, by Anyone
Topology	Cellular	Point-to-point/multipoint, star, cell or mesh tree
Device Types	Base station (gNB), User Equipment (UE)	Radio Device (RD) with context-based roles
Infrastructure	PLMN; NPN (Campus network)	User deployed RDs
Network operation	MNO	User
Spectrum	Spectrum auctioned to MNO; Site licenses	1, 9 GHZ, IMT-Spectrum, Site licenses, SRD/RLAN
Spectrum Management	Planning by MNO; Expert tool	Local, self-organized

B. System and Network Architectures

DECT-2020 NR enables wireless network deployment in point-to-point/multipoint, star, and mesh tree network topologies. Contrary to conventional cellular technologies, DECT-2020 NR does not distinguish between fixed radio stations (e.g. base stations, eNBs, gNBs) and mobile terminals (e.g. UE). Each DECT NR+ radio device (RD) can adapt its role in the network depending on the context of the application. The roles a RD can take are:

- *Fixed Termination point (FT)*, RD_{FT} , the RD acts as the initiator and coordinator of a set of local resources, other RDs may connect and communicate with it;
- *Portable Termination point (PT)*, RD_{PT} , the RD selects a RD_{FT} for communication, associates with it and follows coordination decisions;
- *Combination of FT and PT*, $RD_{FT,PT}$, the RD acts simultaneously as a RD_{FT} and as a RD_{PT} .

A RD_{FT} is also responsible for routing data from its associated (RD_{PT} s) to an Internet connection point, either directly or through another RD_{FT} .

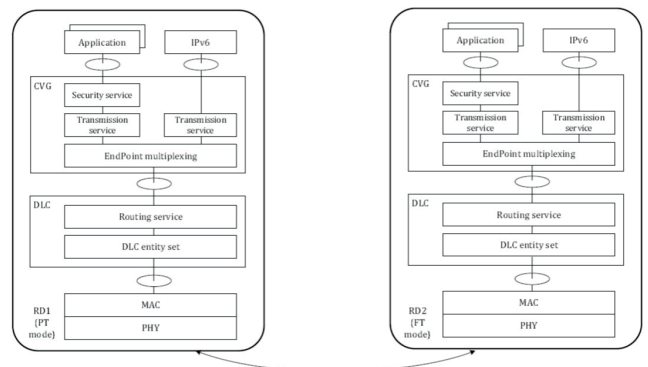


Fig. 2. DECT-2020 NR overall protocol architecture for point-to-point or simplified star topology [6].

URLLC use cases (e.g. cable replacement or wireless local area network) benefit from a point-to-point, point-to-

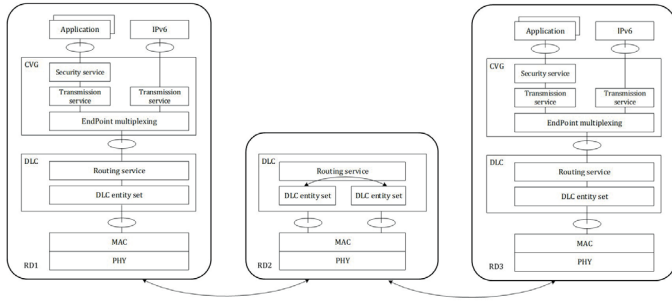


Fig. 3. DECT-2020 NR overall protocol architecture for mesh networking [6].

multipoint, or star topology. For this kind of use cases, the context of the communication tend to be stable, so RDs keep their selected roles during the application lify cycle. Fig.2 illustrates an example of protocol architecture exploiting a point-to-point or a simplified star topology for both IP and non-IP applications. RD1 takes the RD_{PT} and RD2 the RD_{FT} ; changes are not expected.

mMTC use cases benefit from a clustered mesh tree network topology, where RDs may change their role autonomously based on local context so to best contribute to the formation and resilience of the network. Details on the cluster formation and autonomous routing strategies can be found in [6]. Basically, each RD selects the next hop individually among the set of available routes towards the Internet connection point. Routing decisions are taken autonomously by each RD using a cost value without the need to maintain routing tables in each device. The route cost calculation is not standarized but left to implementation. Route cost value will increase at least in 1 unit with every hop. The maximum route cost value is specified to be 254, which is but not seen as a limiting factor in real deployments. Fig.3 illustrates an example of protocol architecture exploiting mesh topology and autonomous routing for IP and non-IP applications.

C. PHY Layer

The PHY layer of DECT-2020 NR addresses in principle all frequency bands below 6 GHz and is described in [7].

The physical layer uses Cyclic Prefix Orthogonal Frequency Division Multiplexing (CP-OFDM) combined with Time Division Multiplexing (TDMA) and Frequency Division Multiplexing (FDMA) according to the Time Division Duplexing (TDD) method. The physical layer numerology offers multiple options for the combination of subcarrier spacing, cyclic prefix length and FFT size, allowing operation with different channel bandwidths and enabling operation in different frequency bands and propagation environments.

Depending on the subcarrier scaling factor $\mu = 1, 2, 4$ or 8 , the OFDM subcarrier spacing can be configured between 27 kHz, 54 kHz, 108 kHz or 216 kHz, respectively. In addition, the Fourier transform scaling factor, β , can be set to allow different channel transmission bandwidth for each configuration of the subcarrier spacing. The minimum nominal channel

bandwidth of 1.728MHz is achieved with the configuration $(\mu, \beta)=(1, 1)$, while the configuration $(\mu, \beta)=(8, 16)$ achieves the maximum channel bandwidth of 221.184 MHz.

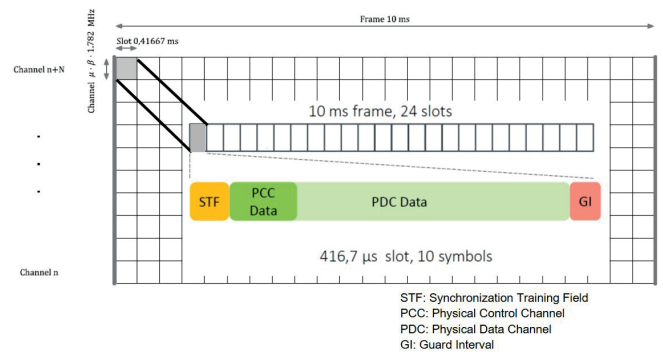


Fig. 4. DECT-2020 NR frame structure [7].

The DECT-2020 NR radio frame (see Fig. 4) has a total duration of 10 ms and consists of 24 slots with a slot duration of $416.67\mu s$. Both frame duration and slot duration have been selected to enable coexistence with legacy DECT systems. Depending on the subcarrier scaling factor, μ , each slot consists of 10, 20, 40 or 80 OFDM symbols. To allow even shorter transmissions, slots can be further divided into 2, 4, 8 or 16 subslots, again depending on the value of μ . It is noted that due to this partitioning, a subslot is always 5 OFDM symbols long, which is also the smallest possible transmission length. The duration of a packet transmission, uplink or downlink, is specified in terms of slots or subslots, with a maximum of 16, respectively.

Further key features are the support for advanced channel coding (turbo coding) schemes for both control and physical channels and Hybrid ARQ (HARQ) with incremental redundancy. The advanced channel coding together with HARQ ensures low latency and highly reliable communication. In addition, the physical layer supports fast link adaptation, transmitter and receiver diversity, and MIMO operation with up to 8 streams.

D. MAC Layer

The overall MAC design of DECT-2020 NR is still ongoing. Initial considerations and fundamental design decisions are described in [8].

The MAC layer provides data transfer and radio resource allocation services to upper layers. The data transfer service can be configured in two operational modes:

- *scheduled access*: for synchronous, periodic URLLC traffic,
- *unscheduled/random access*: for asynchronous immediate packet data transmission

Packets for scheduled services are allocated by the RD_{FT} and are only usable after a MAC signalling process (with confirmed setup and release). On the other hand, packets for random access services can be used without feedback from the

receiving part. Different packet formats have been designed for random access and scheduled access services.

Radio resource allocation functionality is controlled by radio devices operating as RD_{FT} for any scheduled data transfer service. Resources used by scheduled services may be designated as protected so to prevent collisions from non-scheduled access in the same system.

Effort has been taken so that DECT-2020 NR inherits legacy DECT feature of automatic interference management, which eliminates the need for extensive frequency planning. In legacy DECT this feature is called the DECT Channel Allocation/Selection (DCA/S) algorithm and relies on the exchange of "blind slots" information indicating which slots are busy/blind or free. The "blind slots" information in legacy DECT is a bit mask; details on the DCA/S algorithm can be found in [9]. DECT-2020 NR cannot handle resource allocation information exchange by a simple bit mask but it uses a broadcast mechanism over a pilot bearer.

E. Spectrum Bands

DECT-2020 NR is designed for operation in IMT spectrum. The radio channel numbering scheme enables to assign channels from 450 MHz up to 5 875 MHz organized into 17 different operating bands listed in [10]. The nominal RF bandwidth can vary from 1,728 MHz up to 221,184 MHz.

Current release 1 assumes access to legacy DECT spectrum [1 880 -1 900/1 920] MHz and future additional access to new spectrum allocations under 6 GHz. Following candidate spectrum bands have been taken into account in the technical specifications:

- Frequency bands currently allocated to IMT-FT service, i.e. to legacy DECT as a member of the IMT-2000 family (1 900 MHz - 1 980 MHz and 2 010 MHz - 2 025 MHz) [11],
- New license exempt frequencies at the 5 GHz band (SRD/RLAN bands),
- New local area licensed frequencies at midbands (e.g. 3,7 GHz - 3,8 GHz band in Germany, 3,8 - 4,2 GHz in UK).

Operation over higher frequency bands (e.g. 24,25 GHz) is identified as possible in the future, however it is assumed to require a revision of the MAC layer, e.g. with additional subcarrier spacing scaling factors.

IV. AREAS WHERE DECT NR+ CAN BRING VALUE

Innovations in IT and wireless technology are essential elements of the ongoing digital transformation process in vertical industries. Industrial applications often benefit from technological developments in consumer market applications, but have both specific technical and non-technical requirements. While for consumer market application it is fine to provision connectivity features such as data rate, latency, jitter or reliability as statistical averages, many vertical industry applications require provisioning within exact fixed limits. Resilience to disruptions and to limited infrastructure availability, the ability to operate "anywhere, anytime and by anyone", data

sovereignty, network transparency and economic independence in the solution are often further essential requirements.

Policy discussions and technical developments around 5G and beyond have acted and will continue to act as a catalyst for the digitization of industry. However, there is no "one size fits all" 5G solution that can meet the diverse needs, requirements and price points of vertical industry applications. Meanwhile, it has been accepted that a large-scale roll out of 5G, requires tailored, optimized solutions.

Industrial areas that can benefit most from novel DECT NR+ wireless connectivity proposition are identified as:

- for mMTC type of connectivity: manufacturing plants, agriculture, smart cities, energy management in buildings, warehouse logistics, mission critical communication services and the management of distributed energy sources that form a smart grid;
- for URLLC type of connectivity: production (goods, media content) and mission-critical applications.

Following two subclauses list example use cases and applications where DECT NR+ is expected to provide potential for innovation.

A. mMTC Use Cases and applications

In the area of mMTC, DECT NR+ is suitable for e.g.:

- Connectivity of Battery-Powered Mobile Robotic Systems like Autonomous Mobile Robots (AMRs)
- Asset Tracking of Tools, Pallets, Containers, and any other Equipment
- Telemetry of Fixed and Mobile Sensor Units (e.g. Electricity smart metering, Integration of Distributed Energy Resources or Environmental sensing in Cities at larger scale)
- Condition-Based Monitoring, such as Occupancy Detection in Buildings or Machine level Condition monitoring
- Large area environmental monitoring such as early warning systems for Wildfires, heavy flooding or monitoring systems in the crop and cattle growth
- Mission critical communication for the organization and coordination of rescue services in case of disasters

These use cases benefit from DECT NR+'s ability to deploy scalable wireless networks at an efficient cost of ownership based on inherent self-operation and realization of private networks in dedicated spectrum.

Initial application developments are underway in smart metering, where a variety of connectivity standards have been deployed in different regions to date. The combination of long range meshing between nodes, near infrastructure-free networks, and high density in free and dedicated spectrum makes smart power metering an excellent example where DECT NR+ helps overcome known and transparent risks such as high device density, the need for easy deployment and full coverage, and regulatory constraints in the sub-GHz bands.

B. URLLC Use Cases and applications

In the area of URLLC, DECT NR+ is suitable for e.g.:

- Factory automation with deterministic cyclic-traffic requirements
- Professional media content production
- Unified Communications for enterprises: high-quality audio, messaging (alarm, control), positioning
- Healthcare (institutional and home)
- Mission critical connectivity islands

These use cases benefit from DECT NR+'s ability to easily set up and operate self-organizing, low latency, locally confined wireless networks. Beyond, DECT NR+'s decentralized setup and operation makes it ideal for mobile ad-hoc deployments, which is of particular interest for professional media content production [12] and mission critical use cases [13].

Initial application developments are underway e.g. in audio-PMSE, where DECT NR+ can deliver the strict performance (e.g. latency, jitter, synchronization, reliability) required for many professional audio systems (wireless microphones, in-ear systems, intercom) used by touring bands, recording studios, theatres and broadcasting (including electronic news gathering). The use of DECT NR+ for professional audio systems can help meet the ever-increasing demand for spectrum access in all areas of live & recorded entertainment and media streaming.

V. SPECTRUM POLICY AND GOVERNANCE MODELS

Network slicing in public networks and the deployment and operation of customized non-public networks/campus networks by or for industrial companies are initial answers to the special requirements of vertical industries. At the local network level in particular, 5G can bring disruptive change and create new growth opportunities for vertical industries, provided the remaining regulatory challenges related to local spectrum availability can be resolved.

DECT-2020 NR extends by design the radio capabilities of legacy DECT technology to operate in all TDD IMT bands below 6 GHz. The opening of additional bands beyond the DECT core band, e.g. 2.4 GHz, 5 GHz, 5.8 GHz, and 6 GHz bands can offer considerable opportunities for sustainable, independent or cooperative innovations in the area of non-public 5G networks.

To accommodate different application requirements, it may be necessary to consider the operation of DECT NR+ under different spectrum authorization regimes, from local licensing models to emerging spectrum sharing frameworks, such as e.g. LSA/eLSA [14], [15] in Europe or CBRS [16] in the United States. Further, licensing models should be suitable for different operational needs of industrial use cases, including permanent vs. temporary, stationary vs. nomadic.

ETSI TC DECT has recently approved a new work item to study radio spectrum and related technical requirements for the evolution of DECT-2020 NR [17]. This study may include practical evaluation of additional frequency ranges.

Because DECT-NR+ is clearly a complementary technology to 3GPP's eco-system, ETSI TC DECT aim to define appropriate interworking profiles for 3GPP 5G NR. Work is currently

undergoing at ETSI TC DECT for the definition of suitable interworking profiles in [18].

Basically, a DECT-2020 NR network can provide stand-alone non-3GPP access to a 3GPP core network. Therefore, 3GPP foresees two possibilities. The DECT-2020 NR network can access the 3GPP 5G core network either as an untrusted access network via the on-3GPP Inter-Working-Function (N3IWF) or as a trusted access network via a Trusted Non-3GPP Gateway Function (TNGF). For details on the 3GPP 5G core network architecture and the mentioned interworking functionality refer to [19].

Other scenarios of interest are: (i) a RD_{PT} acting as an intelligent gateway between a DECT-2020 NR and a 3GPP network for extended connectivity; (ii) a dual radio interface RD_{PT} , i.e. a radio device that supports both the 3GPP NR and the DECT-2020 RITs, could act as a UE in the 3GPP network while keeping a parallel, on-demand connection to a stand-alone DECT-2020 NR network.

VI. CONCLUSIONS

DECT NR+ is a young radio standard developed by ETSI and DECT Forum. This paper reveals the fundamentals of the technology, highlights relevant use cases and proposes future work directions in the areas of spectrum policy and governance models with the goal to motivate and incentivate research activities.

Key technology features of DECT-2020 NR include advanced OFDM-based physical layer numerology and medium access control algorithms, decentralized setup and operation enabling self-organizing and self-healing wireless networks. Technology is easy to introduce into new IMT frequency bands beyond the legacy DECT core band.

DECT-2020 NR offers local deployment capabilities without the need for separate network infrastructure or network planning. Further, minimal deployment or maintenance requirements make this technology attractive, especially for small companies and medium-sized industries.

The flexible and autonomous operation of the network allows industry players to focus on developing applications and solutions, keeping them completely under their control, without the value of the application being compromised by the additional cost and complexity of network commissioning.

As a pure ETSI standard, it might not be exaggerated to claim that DECT-2020 NR can be one most important innovation driver for more technological sovereignty in Europe.

REFERENCES

- [1] ETSI Evaluation Group; Final Evaluation Report on DECT-2020 NR, ETSI TR 103 810 V1.1.1 (2021-11), November 2021.
- [2] M. Penner, M. Nabeel and J. Peissig, "URLLC Performance Evaluation of IMT-2020 Candidate Technology: DECT-2020 New Radio," 2021 IEEE 94th Vehicular Technology Conference (VTC2021-Fall), 2021, pp. 1-7, doi: 10.1109/VTC2021-Fall52928.2021.9625340.
- [3] Anttonen, A., Karhula, P., Lasanen, M., & Majanen, M. (2021). Enabling Massive Machine Type Communications with DECT-2020 Standard: A System-Level Performance Study. VTT Technical Research Centre of Finland. VTT Research Report No. VTT-R-00367-2.

- [4] Rec. ITU-R M.2150-1, Detailed specifications of the terrestrial radio interfaces of International Mobile Telecommunications-2020 (IMT-2020), February 2022.
- [5] ERC/DEC/(94)03, ERC Decision of 24th of October 1994, on the frequency band to be designated for the coordinated introduction of the Digital European Cordless Telecommunications system.
- [6] ETSI TS 103 636-1: DECT-2020 New Radio (NR); Part 1: Overview
- [7] ETSI TS 103 636-3: DECT-2020 New Radio (NR); Part 3: Physical layer
- [8] ETSI TS 103 636-4: DECT-2020 New Radio (NR); Part 4: Medium Access Control layer
- [9] ETSI EN 300 175-3: "Digital Enhanced Cordless Telecommunications (DECT); Common Interface (CI); Part 3: Medium Access Control (MAC) layer".
- [10] ETSI TS 103 636-2: "DECT-2020 New Radio (NR); Part 2: Radio Reception and Transmission requirements"
- [11] ETSI EN 301 908-10 V4.2.2 (2016-11); Electromagnetic compatibility and Radio spectrum Matters (ERM); Base Stations (BS), Repeaters and User Equipment (UE) for IMT-2000 Third-Generation cellular networks; Part 10: Harmonised Standard for IMT-2000, FDMA/TDMA (DECT) covering the essential requirements of article 3.2 of the Directive 2014/53/EU
- [12] CEPT ECC Report 323; Spectrum use and future spectrum requirements for PMSE; February 2021.
- [13] EU Project FUDGE-5G; <https://fudge-5g.eu/en/use-cases/ppdr>; last visited April 2022.
- [14] ECC Report 205 - Licensed Shared Access (LSA).
- [15] ETSI TS 103 652-1 V1.1.1. (2019-02): Reconfigurable radio systems (RRS); evolved Licensed Shared Access (eLSA); Part 1: System requirements.
- [16] CBRS Final FCC Report & Order; October 2018.
- [17] Study on radio spectrum and related technical requirements for DECT-2020 NR (DTR/DECT-00389)
- [18] ETSI TR 103 670: "DECT Study on Interworking of DECT with 3GPP network" (under development)
- [19] ETSI TS 123 501: 5G; System architecture for the 5G System (5GS) (3GPP TS 23.501 Release 16)